

Tejal Anavekar

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https://tejalanavekar.github.io/Portfolio-Website/

EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Computer Science

Aug 2024 – May 2026

- Relevant skills: Analysis of Algorithm, Database management, Applied Natural Language Processing, Information Retrieval, Geospatial Information Management

Fr.C.Rodrigues Institute of Technology

Mumbai, India

Bachelor of Engineering in Information Technology

July 2020 – June 2024

- Relevant skills: Operating Systems, Computer Networks, Artificial Intelligence, Big Data, Cloud Computing

TECHNICAL SKILLS

Languages: Python, C, C#, HTML, CSS

Engineering Practices: Microservices Architecture, REST API Design, System Design, OOP

Frameworks: React.js, Node.js, Express.js, ASP.NET Core, REST APIs, Minimal API, Flask

Libraries: LangChain, PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy

Databases: PostgreSQL, MongoDB, Firebase, SQL Server (T-SQL)

Cloud & Tools: AWS (EC2, S3), GCP, Linux/Unix, Shell Scripting

EXPERIENCE

Software Developer

Sept 2024 – Present

USC Institute for Creative Technologies

Los Angeles, CA

- Engineered full-stack internal platforms using C# ASP.NET Core Razor Pages, wiring Minimal API endpoints into page handlers for efficient and structured payload delivery across internal services with scalable backend architecture
- Architected normalized SQL Server schemas with FK constraints and stored procedures, leveraging EF Core DbContext and LINQ pipelines to drive dynamic page rendering and automated Excel report generation
- Deployed real-time KPI monitoring dashboards backed by EF Core aggregation queries, validated end-to-end in live ICT production environment with low-latency high-frequency data refresh pipelines

Data Engineer Intern

June 2025 – Aug 2025

Meta Platforms

Burlingame, CA

- Engineered 4 production-grade distributed data pipelines on Meta's internal infrastructure, processing billions of records daily for real-time analytics at petabyte scale via parallelized batch ingestion
- Developed Python-based data validation frameworks enforcing schema integrity, null-safety, and idempotency guarantees and fault-tolerant error recovery across multi-stage pipeline DAGs in high throughput systems
- Optimized DAG-based workflows through query planning and parallelism; solutions adopted by 3+ cross-functional teams

AI Engineer

June 2022 – Apr 2023

Tata Institute of Fundamental Research

Mumbai, India

- Built CV pipelines integrating SIFT feature extraction, improving classification precision by 25% over CNN baselines
- Fine-tuned and deployed ResNet-50 and VGG variants, reducing inference latency by 30% in high-throughput workloads
- Scaled modular preprocessing pipelines like resolution normalization, histogram equalization, brightness calibration for robust, standardized input handling across heterogeneous datasets in production-grade ML pipelines

PROJECTS

Budget Buddy | MERN, LangChain, REST API

Jan 2025 – Present

- Architected modular full-stack application with service-layer separation decoupling API service modules, business logic, and UI components into independently maintainable units enabling scalable RESTful API design and request handling
- Enabled JWT authentication with bcrypt hashing and optimized session handling using Redis caching strategies
- Integrated LangChain-based LLM pipelines for AI-driven financial insights, improving response accuracy by 25%

Gene Semantic Bridge | Pytorch, CVAE, BioBert, scRNA-seq, Generative Modeling

Jan 2026 – May 2026

- Designed KA-CVAE (Conditional VAE) integrating BioBERT embeddings to model unseen CRISPR perturbations
- Engineered parameterized architecture with 128-dim latent space modeling across single and combinatorial perturbations
- Evaluated on Norman scRNA-seq OOD benchmarks, achieving R-squared value of 0.996, outperforming baseline models

Query Bot | NLP, BERT, LSTM, Pytorch

Jan 2025 – Apr 2025

- Implemented seq2seq architecture with fine-tuned BERT encoder and attention-based LSTM decoder over relational schemas
- Enhanced schema-aware cross-attention over DB metadata, improving multi-table query generation and syntactic validity
- Achieved 91.8% execution accuracy and 82.2% exact match on Spider benchmark, outperforming seq2seq baselines

GeoForecast - Weather App | MERN Stack, Flask, Swift, GCP

Aug 2024 – Dec 2024

- Architected microservices-based system decoupling geolocation, weather, and user services into deployable modules
- Deployed containerized APIs on GCP Cloud with autoscaling, achieving 93% location accuracy, and +35% user retention
- Integrated geolocation and weather REST APIs with caching for low-latency real-time and daily forecast delivery